

Reza Badiei

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PROFESSIONAL PROFILE

Electronics Engineer at ASHA Beam Profile with an M.Sc. in Electrical Engineering, specializing in Digital Electronic Systems. Experience spans custom SoC interface PCB design, microcontroller firmware, embedded Linux and camera systems, FPGA development, and CUDA-based computing. Research experience includes developing energy- and resource-efficient hardware for spiking neural networks, including modified neuron models and FPGA implementations. Skilled in using VHDL and Python for modeling, simulation, and analysis, alongside Xilinx development tools.

RESEARCH INTERESTS

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- Digital Circuit Design
 - Artificial intelligence (AI)
 - Approximate computing
 - Internet of Things (IoT)
 - Neuromorphic computing
 - Spiking Neural Networks (SNNs)

EDUCATION

Ph.D. in Electrical Engineering | October 2025 – Present
Sharif University of Technology, Tehran, Iran

M.Sc. in Electrical Engineering, Specialization in Digital Electronic Systems | October 2022 – February 2025
Shahid Beheshti University, Tehran, Iran
Thesis: Designing Approximate Computing Circuits for Developing Spiking Neural Networks
Supervisor: Dr. Somayeh Timarchi
Cumulative GPA: 17.99 out of 20 (89.95%)

B.Sc. in Electrical Engineering | October 2018 – September 2022
Shahid Beheshti University, Tehran, Iran
Thesis: Design and implementation of a complete smart IoT system for monitoring and controlling laboratory events
Supervisor: Dr. Hadi Asharioun
Cumulative GPA: 17.84 out of 20 (89.2%)
Rank: 3rd in program

PUBLICATIONS

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- A. Zakaleh, **R. Badiei**, Z. H. Zadeh and S. Timarchi, "Low-Power Hyperbolic CORDIC Design of the Fitzhugh-Nagumo Neuron for Neuromorphic Systems," *IEEE Embedded Systems Letters*, accepted for publication, 2026, doi: 10.1109/LES.2026.3710888.
 - **R. Badiei**, S. Timarchi and A. Zakaleh, "Low-Power Resource-Efficient FPGA Implementation of Modified FitzHugh–Nagumo Neuron for Spiking Neural Networks," *IEEE Transactions on Circuits and Systems II: Express Briefs*, vol. 72, no. 11, pp. 1780-1784, Nov. 2025, doi: 10.1109/TCSII.2025.3615935.
 - A. Hajjafari, **R. Badiei**, H. Asharioun and R. Nasiri, "Design and Implementation of an IoT-based Smart Home with the Ability to Communicate with the Smart Grid," in *2023 International Conference on Protection and Automation of Power Systems (IPAPS)*, Tehran, Iran, Islamic Republic of, 2023, pp. 1-5, doi: 10.1109/IPAPS58344.2023.10123324.

ACADEMIC SERVICE

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- Reviewer for Neurocomputing (Elsevier)

EXPERIENCE

Research

2023 – 2026 | Research Assistant — Shahid Beheshti University, Tehran, Iran

Conducted research on neuromorphic computing and Spiking Neural Networks (SNNs) under the supervision of Professor Somayeh Timarchi, focusing on energy- and resource-efficient hardware implementations.

- Designed and simulated SNNs for classification with STDP learning.
- Developed modified versions of Izhikevich, FitzHugh-Nagumo, and Hindmarsh-Rose models to improve energy and resource efficiency.
- Conducted experiments to optimize energy and resource usage in neuromorphic hardware.
- Used VHDL for circuit modeling and Python for simulation and analysis.
- Contributed to research papers on neuromorphic computing.

2022 | Electrical Engineer (IoT Designer & Implementer), Research Assistant — Shahid Beheshti University, Tehran, Iran

Developed a smart IoT system for monitoring and controlling lab events as part of my bachelor's project. Integrated diverse devices through a custom communication network.

- Developed communication protocols for seamless device integration.
- Built a real-time IoT dashboard accessible locally and globally.

Teaching

2022 | Teaching Assistant, Electrical Engineering Course — Shahid Beheshti University, Tehran, Iran

Served as a Teaching Assistant for the C++ programming course under Professor Hadi Asharioun. Developed instructional materials and provided support to students to enhance their programming skills.

- Assisted in explaining complex programming concepts and fostering collaborative learning.
- Guided students in C and C++ coding, emphasizing logic and flowchart integration.
- Helped students master syntax and practical coding applications.

Work

May 2026 – Present | Electronics Engineer — ASHA Beam Profile, Tehran, Iran

Work across custom PCB design, microcontroller firmware, embedded SoCs, CUDA-based GPU computing, and FPGA development. Hardware platforms vary according to each task's processing requirements.

- Designed a Cubie A7S SoC interface daughterboard with an STM32G031 for camera and laser triggering; developed STM32 and ESP32 firmware.
- Developed a Linux MIPI camera driver for the Allwinner A733, supporting MV SC130 and VEYE cameras.
- Integrated Hikrobot USB 3.0 and VEYE MIPI cameras simultaneously on the SoC.
- Configured V4L2 and Device Tree and adapted Linux camera drivers for Raspberry Pi 3 and Cubie A7S platforms.
- Used CUDA on NVIDIA Jetson to parallelize calculations for laser profile measurement.
- Developed a Qt application for camera-register configuration, image acquisition, processing, and visualization.
- Worked on FPGA-based system development, hardware implementation, and hardware/software integration.

Aug 2025 – Apr 2026 | Medical Equipment Supplier — Behbud Darman, Tehran, Iran

Worked in the supply and sale of medical equipment, including customer communication, product presentation, and sales activities.

2021 – 2023 | IoT Designer & Implementer — Smart Home Venture, Tehran, Iran

Developed and deployed an IoT-based smart-home prototype as an entrepreneurial venture, working across embedded hardware, wireless communication, and full-stack software.

- Designed and implemented embedded systems for smart devices, including sensor interfacing, PCB layout, and microcontroller programming.
- Developed a full-stack web application for real-time control and monitoring of smart devices using React.js and Node.js.
- Integrated hardware and software components using MQTT and TCP/IP for remote monitoring and automation.
- Collaborated with a small engineering team throughout the prototyping and deployment phases.

Jun 2021 – Nov 2021 | Electrical Engineer Intern — Andishe Negar Pars Company, Tehran, Iran

Collaborated in the development of smart home devices, including PCB design and programming for smart switches, temperature controllers, power outlets, and sensors (motion, gas, and flame detectors). Worked with the company's smart home system protocols to integrate devices seamlessly.

SKILLS

Programming Languages: C, C++, Assembly, Python, VHDL, MATLAB
Embedded Systems & IoT: ARM, AVR, ESP32, ESP8266, STM32, STM8, Arduino; UART, SPI, I²C, DMA, TCP/IP, MQTT
FPGA Design: Xilinx ISE, Vivado Design Suite, Vivado HLS, Xilinx SDK, ModelSim
Embedded Linux & GPU Computing: Linux driver development, V4L2, Device Tree, CUDA, NVIDIA Jetson
Design & Development Tools: Altium Designer, STM32CubeIDE, CodeVisionIDE, Arduino IDE, Qt Creator, Multisim, Proteus
AI & Software: TensorFlow, PyTorch, Node-RED, Linux shell scripting, HTML, CSS, JavaScript, React.js, TypeScript, Next.js, Node.js, Express.js, SQL, Nginx

SELECTED PROJECTS

- Designed and simulated a Human Activity Recognition system in a smart home environment using deep learning models such as CNN and LSTM in “Deep Learning course”. [June 2023]
- Simulated and implemented the paper "Hardware/Software Co-Design of a Dynamically Configurable SHA-3 System-on-Chip (SoC)" in “ASIC/FPGA Circuits course”. [June 2023]
- Designed and implemented a Direct Digital Synthesis (DDS) interface for the AD9951 IC using VHDL in “VHDL course”. [February 2023]
- Designed and implemented a voice recognition system using Gaussian Mixture Models (GMM) for smart access control in “Machine Learning course”. [June 2022]

RELEVANT COURSEWORK

• VHDL	19.12/20	• Interface Circuits	19.5/20	• Linear ICs Design	19.5/20
• ASIC/FPGA Circuits	20/20	• Internet of Things	20/20	• Electronics II	20/20
• Advanced Digital Electronics	18.25/20	• Machine Learning	19.5/20	• Electronics III	19.9/20

ENGLISH LANGUAGE PROFICIENCY

- **Level:** Upper-Intermediate
- **IELTS Academic:** Overall Band 6.5 (August 2025)